

EXECUTIVE TECHNICAL ORIENTATION

Connection, Control & Execution Authority

Prepared in response to David G. Holmberg's October 2, 2026 technical challenge

THE QUESTION TA-14 GOVERNS

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Purpose of this orientation

David, you have seen only a narrow surface of TA-14. This document answers your questions directly, maps them to the pre-existing architecture, incorporates the subsequent technical clarification surface, identifies what your challenge clarifies, and marks the boundary where TA-14 deliberately stops rather than claiming the work of engineering, safety, cybersecurity, operations, regulation, or hazard analysis.

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Executive orientation - working technical response; not certification, regulatory approval, engineering instruction, production validation, adoption, or endorsement.

1. Executive answer

Your follow-up identifies real operational failure modes: a command can be valid in transport and still be wrong in consequence; safeguards should reflect consequence severity; hazard review may need a wider field of view; humans, warnings, interlocks, sensors and procedures may matter; and systems, assumptions and authority degrade or change over time.

TA-14's present architectural response: those conditions belong in the proof burden for a proposed consequence when the applicable domain requires them. TA-14 does not invent the underlying engineering requirement. It governs whether the applicable requirement is sufficiently evidenced, current, authoritative, bound and supported by standing before commit and execution.

The three-layer distinction

LAYER	QUESTION
Connection / control	Can the message or command travel? Is the source allowed? Is the destination allowed? Is the packet valid?
Domain / safety	What safeguards, limits, interlocks, procedures, training, redundancy or hazard controls are required for this action?
TA-14 consequence	For this exact proposed consequence, are the required evidence, authority and standing sufficient NOW, and do they survive binding and commit?

**CONTROL RIGHT != CONSEQUENCE AUTHORITY != COMMIT PERMISSION
CONNECTION != CONSEQUENCE · TRACEABILITY != TRUTH**

What your challenge changes

It does not justify expanding TA-14 until it owns every technical discipline. It does justify stronger fixtures, clearer teaching, sharper terminology and explicit mapping from domain requirements into the admissibility burden.

2. Your questions, answered directly

Can a Connection Profile grant control capability and thereby grant execution authority?

A Profile may establish bounded exchange or control rights and may carry authority context. A valid delegated grant may satisfy Applicable Authority when the grantor actually holds that authority and the grant remains applicable to the exact consequence. Connectivity, authentication and technical capability are not, by themselves, consequence authority or commit permission.

What if data is wrong, stale, null, conflicting, or produced by a failed sensor?

Then the evidence burden may fail. TA-14 separates record formation, continuity, evidence governance and admissibility so that the existence of a value is not equivalent to sufficient evidence for the proposed consequence.

Where does provenance live?

Source devices, controllers, networks and custodians create and preserve provenance. TA-14 evaluates what is presented; it does not manufacture provenance or make traced data true.

When is no Connection Profile needed?

TA-14 does not define CNS/CP applicability. If a workflow does not cross the connection boundary being governed, the TA-14 consequence burden may still exist even though the implementation route is entirely local.

If a CMMS receives BAS status and returns GO / NO-GO, where is the decision?

A recommendation is not automatically the commit. TA-14 asks where the consequence becomes bound, what actor or mechanism has standing there, what evidence is relied upon, and where the hard-to-reverse transition occurs.

What about a technician?

Human action can face the same consequence examination. But unless the disposition is bound to an enforcement mechanism, TA-14 cannot physically stop an unbound human hand.

What about a technical path that bypasses governance?

Cybersecurity and gateway controls protect technical paths. TA-14 governs only the execution path to which its disposition is actually bound; a bypassed path is not silently governed by a label.

3. Challenge disposition: confirmed, clarified, or still open

This matrix is intentionally stricter than an 'already covered' response. It separates compatibility from prior architecture, clarification work, and questions that remain implementation- or domain-dependent.

DAVID'S OBSERVATION	STATUS	TA-14 RESPONSE / BOUNDARY
Safeguards should scale with risk.	CONFIRMED / COMPATIBLE	The consequence determines the proof burden. TA-14 does not prescribe a universal interlock, sensor, or redundancy count.
A wide-angle hazard review may be required.	DOMAIN INPUT	Where applicable authority requires hazard review, its required outputs become part of what must be established. TA-14 is not itself a universal HAZOP or fire-safety method.
Training, warnings, humans and physical interlocks may matter.	REPRESENTABLE	They may contribute evidence, competence, standing, refusal capacity, safety conditions or enforcement. None is universally mandatory merely because TA-14 exists.
Processes and systems degrade; conditions change.	ALREADY EXPLICIT	Living Authority, Continuous Admissibility, continuity, changed-condition revalidation and execution-reality checks address this class of problem.
Remote evidence and authority can be legitimate.	AGREED + CLARIFY	Local means the receiving consequence domain/boundary, not physical co-location.
Generic review guidance.	ACCEPT NARROWLY	TA-14 can guide governance questions and proof burden; it should not claim ownership of every engineering or hazard-analysis method.
Exactly which safeguard is sufficient?	OPEN / DOMAIN-DEPENDENT	Requires applicable engineering, safety, regulatory and operational authority.

Historical discipline: David's challenge remains independent evidence. Map each proposition to publication dates before claiming prior coverage. Any genuine gap is added prospectively under change control - never retroactively inserted into a frozen baseline.

4. David's AHU example - one-page architecture map

Suppose a gateway accepts a valid BACnet command from an allowed source to an allowed AHU. The packet is authentic and syntactically valid. That is useful evidence. It is not, by itself, the whole consequence determination.

1. CONNECTION / CONTROL	2. DOMAIN REQUIREMENTS
Allowed source Allowed destination Valid packet Bounded control right	Hazard analysis Limits / interlocks Training / procedure Redundancy / safety rules

3. TA-14 CONSEQUENCE DETERMINATION

Reality -> Record -> Continuity -> Admissibility -> Binding -> Commit -> Execution -> Outcome

Are the applicable requirements evidenced, current, attributable, authoritative and bound to this exact AHU, command, scope and time?

4. EXECUTION	5. OUTCOME
Command crosses the consequence boundary only under the established envelope.	What actually happened? What evidence supports that claim?

The boundary: If the applicable hazard review says this command requires a physical interlock, redundant sensor, human confirmation, trained technician, warning, or another safeguard, TA-14 does not invent that requirement. TA-14 asks whether the requirement is actually established and remains applicable before consequence.

Why this matters

A packet can pass every connection and cybersecurity check yet still be inadmissible for consequence because the state is stale, a required safeguard is absent, standing has changed, the target is wrong, or the conditions supporting an earlier ALLOW no longer hold.

5. Claim ownership matrix - what establishes what?

This is the strongest clarification added by the technical showroom. It prevents one layer from silently claiming the function of another.

LAYER	CAN ESTABLISH	DOES NOT AUTOMATICALLY ESTABLISH
Authentication / Authorization	Requesting identity, credential validity, transport security, interface access.	Consequence authority; that the action is admissible now.
Authority grantor / source	Consequence authority and bounded delegation of authority it actually holds.	Technical enforcement; authority the grantor does not hold.
Human / requesting actor	Request and stated purpose; for human work, the physical act itself.	Its own authority or standing merely by asking; a technical stop on a human hand.
CNS/CP + instance	Named parties, purpose, permitted exchange, control rights/limits, validity and revocation terms.	Truth of data; provenance; receiving system decision unless explicitly implemented there.
Evidence / custody systems	Source identity, timestamps, transformations, quality flags, custody/access history.	Truth merely because provenance is complete; operational authority.
Safety interlocks	Prevention of defined unsafe states; assigned life-safety sequences.	Authority or permission merely because no interlock is active.
Enforcement / commit point	Actual refusal or write; assigned technical checks; TA-14 disposition when bound to the path.	Governance of paths that bypass it.
TA-14 consequence examination	ALLOW / HOLD / DENY / ESCALATE on sufficient Admissible Evidence, Applicable Authority and Established Standing NOW.	Creation/custody of provenance; truth; hazard analysis; cybersecurity; execution itself.
Physical execution	The act that changes reality.	That the act was admissible; proof of outcome unless preserved.

6. Enforcement: governance or advice?

TA-14 must sit at the governed commit boundary for its ALLOW / HOLD / DENY / ESCALATE disposition to govern execution.

The falsifiable test

If TA-14 returns HOLD but the same proposed consequence can still execute through the governed execution path, TA-14 has not been established as the execution-governance commit boundary for that path. It is advisory there.

What this does not claim

It does not claim TA-14 blocks every compromised credential, alternate BAS front end, manual action, or technical exploit. Cybersecurity, gateway design, interlocks, human procedure and physical enforcement remain the responsibilities of their own systems and authorities.

Local does not mean nearby

David's remote-EMS example exposes a possible misunderstanding of 'execution authority must be established locally.' Here, local means established for the specific consequence in its receiving governing context - not a human, sensor, controller or authority physically beside the equipment. Evidence, operators, computation and valid delegated authority may be remote.

One sentence to carry forward

A governance label is not a technical barrier. A disposition governs execution only where the execution path is actually bound to honor it.

7. The TA-14 architecture behind these answers

The technical showroom exposes only a small surface. This is the shortest relevant map of the published architecture.

ARCHITECTURE	WHY DAVID SHOULD CARE	RELEVANCE
Admissible Execution Architecture (AEA)	Parent architecture	Evidence-first governance of consequence-bearing execution; establishes the eight-anchor chain.
Evidence Governance + Admissible Evidence	Upstream evidence	Separates a stored value or assertion from governed/admissible evidence capable of supporting consequence.
Admissible Continuity Architecture	Chain survival	Tests whether a record remains a governed witness across time, custody, transformation and reliance.
Living Authority Standard (LAS)	Authority lifecycle	Authority is not static permission; validity depends on continuing conditions.
Continuous Admissibility Governance (CAG)	Ongoing standing	Addresses preservation/re-establishment as reality, evidence, identity, delegation and conditions evolve.
Authority Passport Protocol (AVP)	Federated context	Carries bounded authority context while preserving a receiving-domain authority question.
Admissible Federation Architecture (AFA)	Cross-domain seam	Authority context may cross independently governed domains; execution authority does not simply travel with it.
Execution Authority Boundary Architecture (EABA)	Commit boundary	Focuses on present, bound execution authority at commit, including changed conditions and evidence continuity.

7. Architecture map - continued

ARCHITECTURE	WHY DAVID SHOULD CARE	RELEVANCE
Building Admissibility Gate (BAG)	Built environment	Applies the parent architecture to building information-to-physical-consequence transitions.
Human Performance Stack (HPS)	Human action	Governs the moment before human action becomes consequence and includes changed-condition revalidation.
Admissible Human Intervention Architecture (AHIA)	Human intervention	Asks whether human intervention itself is admissible rather than assuming human-in-the-loop is sufficient.
<u>Admissible Computation Architecture (ACA)</u>	Computational commitment	Applies governance before computational commitment and treats ALLOW as condition-bound.
Governed Reality Architecture (GRA)	Claim boundary	TA-14 does not manufacture truth, create authority, replace engineering/domain expertise, or guarantee safe outcomes.
Runtime Admissibility Protocol (RAP)	Runtime implementation	Addresses exchange, evaluation, reserve, commit, execute, observe and replay across independent systems.
Replay Verification / IRRS	Independent verification	Supports reconstruction/replay of protected consequence from preserved evidence.
Proof of Restraint	Verified non-execution	Supports evidence-bounded claims that protected consequence was correctly restrained or did not occur.

Important: This orientation summarizes the supplied TA-14 publication inventory. Frozen publications remain controlling for exact normative language, identifiers and scope.

8. What TA-14 does - and deliberately does not do

BOUNDARY IS A FEATURE, NOT A GAP

TA-14 governs the right of a proposed consequence to proceed. It does not become the technical authority for every discipline whose requirements may bear on that consequence.

TA-14 DOES	TA-14 DOES NOT
Ask whether evidence is sufficient for the exact proposed consequence.	Make a sensor true merely because its record is well formed.
Test continuity, binding, standing and changed conditions.	Create legal, regulatory, institutional or professional authority that does not otherwise exist.
Preserve separation between capability, authority and consequence.	Replace engineering, fire protection, cybersecurity, hazard analysis, code compliance or professional judgment.
Require applicable domain conditions to be established when they govern the action.	Invent universal safeguards, interlock counts, redundancy levels or review intervals.
Govern commit where its disposition is bound to the execution path.	Claim governance over a path that can bypass the bound commit point.
Preserve frozen baselines and add genuine discoveries prospectively.	Rewrite historical records so a later challenge appears to have been answered earlier.

9. A short verification path through the TA-14 corpus

You do not need to read the entire corpus to test the claims in this response. These records form the shortest path from the parent proposition to authority lifecycle, federation, commit and changed-condition governance.

PUBLICATION / DATE	WHY READ IT	RELEVANCE TO YOUR QUESTIONS
AEA - Volume 1 Foundational Monograph v1.2 May 30, 2026	START HERE	Parent architecture and core execution chain.
Living Authority Standard (LAS) July 14, 2026	AUTHORITY	Authority is continuously conditioned rather than static permission.
Continuous Admissibility Governance (CAG) July 14, 2026	CHANGE OVER TIME	Preservation/re-establishment as reality, evidence, identity, delegation and conditions evolve.
Authority Passport Protocol (AVP) v1.0.2 July 15, 2026	FEDERATED CONTEXT	Authority context across domains and receiving-boundary treatment.
Admissible Federation Architecture (AFA) v1.0-RC1 September 19, 2026	CROSS-DOMAIN	Protocol-neutral federation seam; tests whether execution authority travels.
Execution Authority Boundary Architecture (EABA) v1.0-RC1 September 20, 2026	COMMIT	Present, bound authority at the point of commit.
Human Performance Stack (HPS) v1.0 September 26, 2026	HUMAN ACTION	Human standing and changed-condition revalidation.
Building Admissibility Gate (BAG) September 3, 2026	BUILDINGS	Building-specific information-to-physical-consequence profile.
Governed Reality Architecture (GRA) August 30, 2026	BOUNDARIES	Explicit claim boundaries; does not replace engineering/domain expertise.
Admissible Computation Architecture (ACA) September 8, 2026	COMPUTATION	Condition-bound admission and changed-condition revalidation research.

Verification note

Publication titles with verified DOI targets in this edition are directly clickable. Where a direct DOI has not been independently verified, no link has been invented. The frozen Zenodo publication remains controlling for exact normative language, identifiers and scope.

10. Break this: eight showroom experiments

These are not demonstrations of correctness. They are tests designed to expose where the architecture is wrong.

#	EXPERIMENT	WHAT IT SHOULD EXPOSE
01	Revoke mid-request	A valid instance revoked before commit must not silently remain executable.
02	Stale sensor	A plausible value can still fail the declared source-age requirement.
03	Control right without standing	A bounded control right does not create standing for the requested consequence.
04	Local safety interlock	Valid control right + delegated authority still cannot override an applicable local safety interlock.
05	Wrong device / good data	Fresh accurate evidence from the wrong physical device must not support the consequence.
06	Read-only recommendation	A recommendation must not silently become a control right.
07	Pre-authorized band	In-range and out-of-range requests must be distinguishable at enforcement.
08	Unauthorized technical path	A compromise outside the governed connection is recorded separately from governed-path refusal.

The write experiment - designed, not yet run

AHU-1 supply-air temperature: 58 F -> 62 F

Requesting party: Org B, contracted optimisation provider.

Control right: Owner <-> Org B; write AHU-1 SAT setpoint 55-65 F, business hours, valid 7 days, revocable.

Consequence authority: written delegation from the building owner.

Evidence: fan status, current SAT, outdoor-air temperature, source device identity and source timestamps.

Safety: freeze-stat and fan-proven interlocks.

Enforcement point: BAS integration gateway on the governed path.

11. The write experiment - falsifiable run matrix

RUN	CONDITION	LAYER TESTED	MUST BE OBSERVED / FALSIFIED IF
1	Everything valid.	All three layers	Gateway writes 62 F; every check and ALLOW recorded. Fails if refused without reason or a check is missing.
2	Grant revoked one second before commit.	Control right / instance	Gateway refuses; no write. Fails if write executes.
3	Freeze-stat / safety interlock active.	Commit permission / safety	Interlock blocks; HOLD; no write. Fails if write executes.
4	Required outdoor-temperature evidence stale.	Admissible Evidence	HOLD because source age exceeds declared limit. Fails if custodian freshness substitutes for source freshness.
5	Requested setpoint 70 F, outside 55-65 F scope.	Control right / scope	Gateway refuses; no write. Fails if write executes or is silently clamped without record.
6	Grant issued by party without authority over AHU-1.	Consequence authority	Authority check fails; DENY or ESCALATE; no write. Fails if grant is treated as authority without checking grantor.
7	TA-14 returns HOLD; same write attempted through BAS front end.	Commit boundary coverage	Refused only if front end is routed through governed boundary. If it executes, TA-14 is advisory on that path.

The point of this experiment is not to prove TA-14 by assertion. It is to make the boundary observable: what was accepted, what was refused, which layer made the determination, whether HOLD actually prevented commit, and whether the record can prove what happened.

12. What we should change because of your challenge

The correct response is not to enlarge TA-14 until it owns hazard analysis. The correct response is to make the existing boundary easier to inspect and harder to misunderstand.

1. Add fixtures where transport/authentication passes but consequence is HOLD or DENY because evidence, standing, state or applicable safeguards fail.
2. Add sensor-disagreement, stale-value, null-value, computation-error, malicious-but-authenticated-command and changed-configuration fixtures.
3. Add a remote-EMS fixture demonstrating that receiving-boundary establishment is governance context, not physical location.
4. Make risk-proportional domain requirements explicit as inputs without prescribing universal safeguards.
5. Preserve David's October 2 email and this subsequent clarification as independent challenge evidence and map each proposition to the frozen publication chronology.
6. Where a genuine gap survives that comparison, incorporate it prospectively under explicit change control.

Closing proposition

Can a consequence-bearing system establish sufficient current evidence, applicable authority and standing for the exact proposed consequence, preserve that determination through binding and commit, and prove what actually happened - without allowing capability, connectivity, prior permission, stale evidence or transported context to silently substitute for present execution authority?

David, thank you for challenging TA-14 at the practical boundary. The useful outcome is not that TA-14 agrees with every proposed extension. The useful outcome is that each challenge can be traced against the frozen record: what was already there, what needs clearer teaching or implementation, what genuinely needs revision, and what properly belongs to another technical authority.

Capability != Authority. Understanding != Permission. Commissioned once != authorized forever.

Source note

This executive orientation is based on David G. Holmberg's October 2, 2026 technical challenge, the TA-14 publication inventory reflected in the supplied source material, and the subsequent public technical clarification surface 'Connection, Control & Commit Permission.' Publication summaries are orientation aids; frozen publications remain controlling for exact normative language. No endorsement by David Holmberg, NIST, or any other participant is asserted.

WORKING TECHNICAL RESPONSE - OPEN FOR CORRECTION